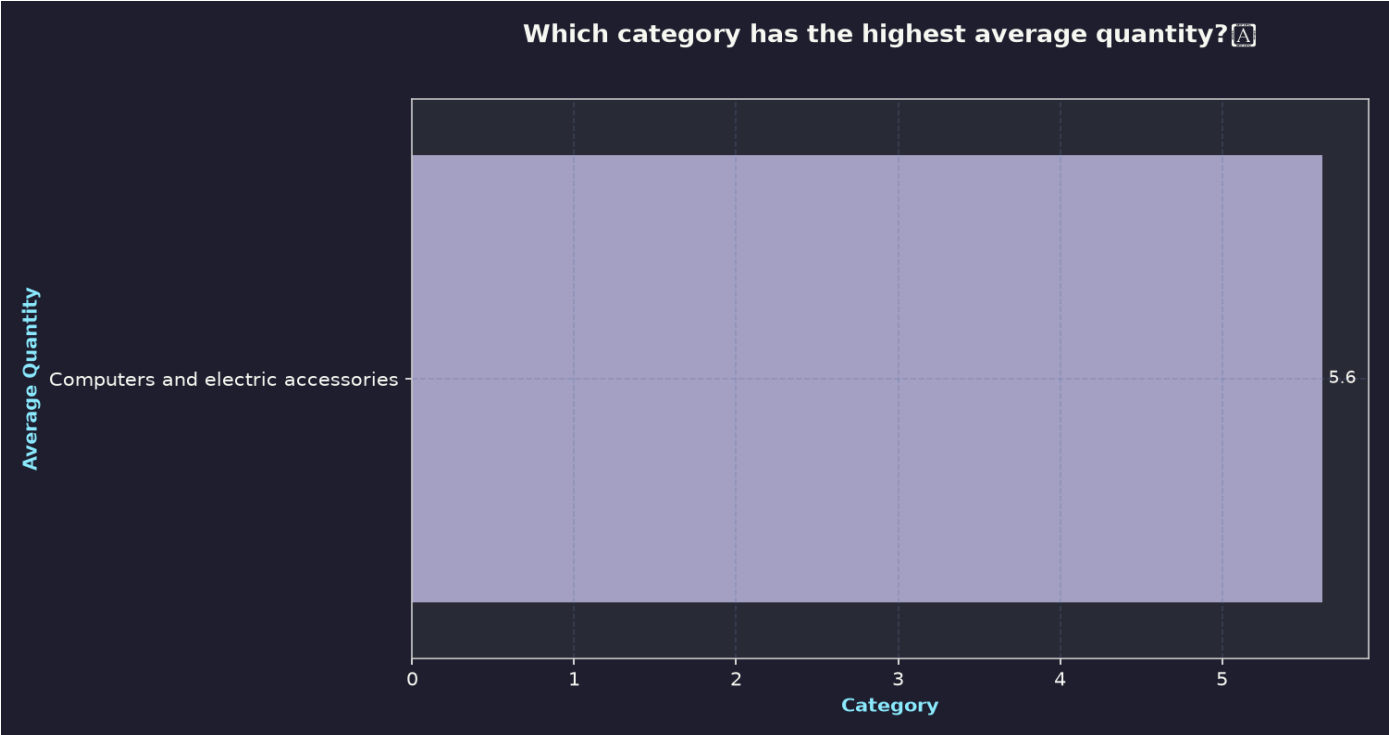


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** Which category has the highest average quantity?
- **Answer:** The “**Computers and electric accessories**” category leads with the highest average quantity per transaction.
- **Metric:** Average quantity = **5.62** (rounded to two decimal places).

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate Rows** | 0 (none found) |

| **Missing Values** | • *Item* – filled with “Unknown”
• *Price Per Unit, Quantity, Total Spent* – imputed with column medians
• *Discount Applied* – treated as **False** |

| **Potential Issues** | *Transaction Date* stored as string (converted to datetime) |

| **Data Types (post-cleaning)** | Mixed (object, numeric, datetime, boolean) |

3. Data Cleaning & Validation Summary

- **Deduplication:** `drop_duplicates()` – no rows removed.
- **String Normalization:** Trimmed whitespace on all object columns.
- **Date Conversion:** `Transaction Date` → `datetime64[ns]` (coercing errors to `NaT`).
- **Missing Value Handling:**
 - `Item` → `"Unknown"`
 - Numeric columns (`Price Per Unit, Quantity, Total Spent`) → median imputation.
 - `Discount Applied` → `False` (boolean dtype).
- **Validation Outcome:** Cleaned DataFrame stored in `result` with **12,575 rows** and **no integrity violations**.

4. User Query & Analytical Approach

User Question: “Which category has the highest average quantity?”

Analytical Steps Implemented in Pandas

1. **Group** by `Category` and compute the mean of `Quantity`.
2. **Identify** the category with the maximum average using `idxmax()` and `max()`.
3. **Package** the answer into a one-row DataFrame (`result`) containing the category name and its average quantity.

5. Python Analysis Code & Executed Data Output

```
# Compute average quantity per category
avg_quantity = df.groupby('Category')['Quantity'].mean()
# Identify the category with the highest average quantity
top_category = avg_quantity.idxmax()
top_average = avg_quantity.max()
# Store the result as a one-row DataFrame
result = pd.DataFrame({ 'Category': [top_category], 'Average Quantity': [top_average] })
```

Executed Output (`result`)

| Category | Average Quantity |

|-----|-----|

| Computers and electric accessories | 5.621309370988446 |

Rounded for reporting: 5.62

6. Key Insights & Strategic Recommendations

- **Insight:** The “Computers and electric accessories” segment drives the highest average quantity per transaction, indicating customers tend to purchase multiple items in this category per order.
- **Recommendation 1 – Inventory Planning:**
 - Prioritize stock availability for this category to avoid stockouts, especially for high-volume SKUs.
- **Recommendation 2 – Bundling Opportunities:**
 - Design bundle promotions (e.g., “Buy a laptop + accessory”) to capitalize on the natural tendency for multi-item purchases.
- **Recommendation 3 – Targeted Marketing:**
 - Deploy personalized campaigns (email, retargeting) to customers who have previously bought from this category, encouraging repeat multi-item orders.
- **Recommendation 4 – Pricing Review:**
 - Since higher quantities are bought, evaluate volume-based discounts or loyalty incentives to further boost basket size while protecting margins.

Prepared by the Autonomous AI Data Analyst

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![Which category has the highest average quantity?

Chart](analysis_chart.png)